CHE 435/525 *Process Systems Analysis and Control*

Lecture 19: **Loop Bode Diagram, Bode Stability Criterion**

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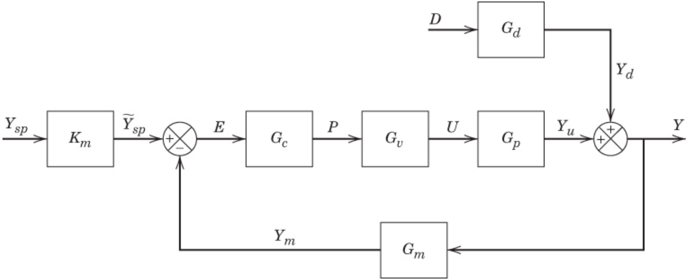
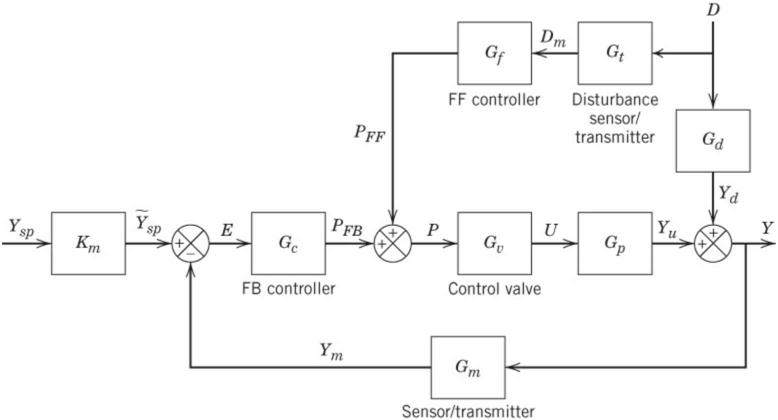
Recapture: Feedback controller synthesis



$$\frac{Y}{Y_{sp}} = \frac{G_c G_v G_p K_m}{1 + G_c G_v G_p G_m}$$

- Feedback controller design $Q = \frac{f}{\tilde{G}^-}$, $G_c = \frac{Q}{1 - Q G_p G_v G_m}$
  - Ideally, synthesize by $Q = 1/G_v G_p K_m$ (targeting at $Y/Y_{sp} = 1$)
  - Actually, subject to limitations in rel. deg., non-min.-phase, and delay

Recapture: Feedforward controller synthesis



$$\frac{Y}{D} = \frac{G_d + G_f G_t G_v G_p}{1 + G_c G_v G_p G_m}$$

- Ideally, synthesize by $G_f = -G_d/G_t G_v G_p$ (targeting at $Y/D = 0$)
- Actually, subject to limitations in rel. deg., non-min.-phase, and delay

Learning objectives of this lecture

1. Plot and sketch the **Bode plot** of the open-loop transfer function G_{OL}
2. Explain the **stability criterion in frequency domain** – Bode criterion
3. Calculate **phase margin and gain margin** for given feedback loop

Controller design in frequency domain: Motivations

- Objectives: **Stability and performance** in closed loop
- The approach we used so far is a time-domain one ...
  - Stability: closed-loop poles (roots to $1 + G_{OL}(s) = 0$)
  - Performance: ISE/ISC from closed-loop TFs $(Y/Y_{sp}, Y/D, U/Y_{sp}, U/D)$
  - Often reduce the problem to one-parameter tuning
- An alternative but more systematic approach: frequency-domain
  - Why?

Controller design in frequency-domain: Motivations

- Every closed-loop t.f. $H(s) = (\cdot)/(1 + G_{OL}(s))$, $G_{OL} = G_c G_v G_p G_m$
- Frequency response (if there is a sinusoidal disturbance or setpoint):
  $$Y/D = G_d(i\omega)/(1 + G_{OL}(i\omega))$$
  We can evaluate amplitude ratio AR and phase angle ϕ under any ω

- Relevance to **performance** ... We hope that CV is not affected much by a sinusoidal disturbance
- Relevance to **stability** ... What if there is an ω with $G_{OL}(i\omega) = -1$?
  - Then closed-loop stability would be lost (A.R. $\to \infty$).
  - The frequency response of G_{OL} is of importance to controller design

Bode diagram of the open-loop TF $G_{OL}(s)$

- Since $G_{OL} = G_c G_v G_p G_m$,

$$\text{AR}_{OL} = \text{AR}_c \times \text{AR}_v \times \text{AR}_p \times \text{AR}_m$$
$$\phi_{OL} = \phi_c + \phi_v + \phi_p + \phi_v$$

- On AR-plot (log-scale) as well on ϕ-plot, add four plots together.

Bode diagram of G_{OL}

Suppose that $G_c = 3$, $G_m = \frac{1}{s+1}$, $G_v = \frac{1}{2s+1}$, $G_p = \frac{2}{5s+1}e^{-1s}$. Plot (sketch) the Bode diagram of G_{OL}.

- Answer:

$$\text{AR} = \left| 3\frac{1}{i\omega + 1}\frac{1}{2i\omega + 1}\frac{2}{5i\omega + 1}e^{-1i\omega} \right| = \frac{6}{\sqrt{(\omega^2 + 1)(4\omega^2 + 1)(25\omega^2 + 1)}}$$
$$\phi = -\arctan\omega - \arctan 2\omega - \arctan 5\omega - 1\omega$$

- Recall: How to sketch? – Shape of each term, and add up

Shape of G_c

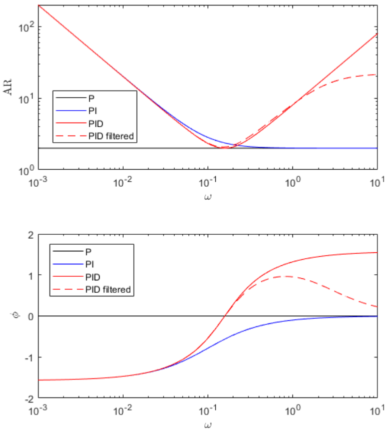
- The Bode diagram of $G_{OL}(s)$ involves the Bode diagram of $G_c(s)$ – to be designed
  - P-controller: $G_c = K_c$. AR $= K_c$, $\phi = 0$ (flat lines)
  - PI-controller: $G_c = K_c(1 + \frac{1}{\tau_I s}) = K_c \frac{1+\tau_I s}{\tau_I s}$

$$\text{AR}_c = \frac{K_c\sqrt{1+\omega^2\tau_I^2}}{\omega\tau_I}$$

$$\phi_c = \arctan \omega\tau - \frac{\pi}{2}$$

  - PID-controller: $G_c = K_c(1 + \frac{1}{\tau_I s} + \tau_D s) = K_c \frac{1+\tau_I s+\tau_I\tau_D s^2}{\tau_I s}$
    - Not easy to sketch

Bode diagram of P, PI, PID controllers



- I mode: At low frequencies, **raises AR by one slope** and **costs $\pi/2$ from ϕ**
- D mode: At high frequencies, **raises AR by one slope** and **gives $\pi/2$ to ϕ**
  - Actually may have to be filtered... AR will plateau, ϕ will return

Controller design in frequency domain

- Recall: Any closed-loop t.f. $= \frac{\cdots}{1+G_{OL}(s)}$!
- **Key observation:** For closed-loop stability, $G_{OL}(i\omega)$ must keep away from "-1" at all frequencies
  - Design G_c so that $G_{OL} = G_c G_p G_v G_m$ is away from -1
- If there was an ω such that $G_{OL}(i\omega) = -1$, then at this frequency, AR $= 1$ and $\phi = -\pi$

Definition

For the open-loop TF $G_{OL}(s)$, define

- Gain crossover frequency ω_g: where AR $= |G_{OL}(i\omega_g)| = 1$;
- Critical frequency ω_c (or phase crossover frequency, ω_p): where $\phi = \angle G_{OL}(i\omega_c) = -\pi$.

Bode stability criterion

Bode stability criterion

Assume that G_{OL} is proper and stable (i.e., the system is open-loop stable), with unique ω_g and ω_c:

$$|G_{OL}(i\omega_g)| = 1, \quad \angle G_{OL}(i\omega_c) = -\pi.$$

Then the closed-loop system is stable if either one of the following holds:

$$\angle G_{OL}(i\omega_g) > -\pi, \quad |G_{OL}(i\omega_c)| < 1.$$

- The proof will need Complex Analysis (MA 513)... let's avoid this.
- Why do we need **another stability criterion** (in addition to "all roots of $1 + G_{OL}(s) = 0$ are on the LHP")?
  - They must be equivalent... indeed, they are.
  - Bode criterion is more convenient (especially with time delays).
  - Bode criterion gives useful robustness concepts (next lecture).

Bode stability criterion: FOPTD Example

P controller for a FOPTD [cf. Seborg Example 14.7]

Suppose that $G_v = G_m = 1$, $G_p = \frac{K_p}{\tau_p s+1}e^{-\theta s}$, with P controller $G_c = K_c$. Determine the range of K_c to ensure closed-loop stability.

- $1 + G_{OL} = 1 + \frac{K_c K_p}{\tau_p s+1}e^{-\theta s}$... the roots can't be explicitly related to K_c!
- Bode criterion: $G_{OL}(i\omega) = \frac{K_c K_p}{\sqrt{1+\omega^2\tau_p^2}}$, $\angle G_{OL}(i\omega) = -\omega\theta - \arctan \omega\tau_p$.
  1. By definition $|G_{OL}(i\omega_g)| = 1$, we get
  $$\omega_g = \frac{1}{\tau_p}\sqrt{K_c^2 K_p^2 - 1}.$$
  2. To ensure $\angle G_{OL}(i\omega_g) > -\pi$, we need
  $$-\omega_g\theta - \arctan \omega_g\tau_p > -\pi$$
  $$\frac{\theta}{\tau_p}\sqrt{K_c^2 K_p^2 - 1} + \arctan \sqrt{K_c^2 K_p^2 - 1} < \pi$$

Bode stability criterion: FOPTD Example

3. ** Simplify: It suffices that $\frac{\theta}{\tau_p}\sqrt{K_c^2 K_p^2 - 1} < \frac{\pi}{2}$, i.e.,

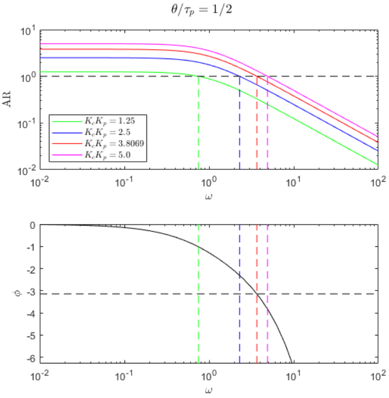
$$K_c < \frac{1}{K_p}\sqrt{1 + \left(\frac{\pi\tau_p}{2\theta}\right)^2}.$$

Further simplify ... it suffices that

$$K_c < \frac{1}{K_p}\frac{\pi\tau_p}{2\theta}$$

- A rule about process control with delay
  1. Process delay (vis-à-vis process time constant) requires that the controller gain **should not be too high**
  2. Upper bound on the controller gain $K_{cu} \propto (\theta/\tau_p)^{-1}$

Bode stability criterion: FOPTD Example



- Looking at the Bode plots ($K_p = 1$, $\tau_p = 1$, $\theta = 1/2$) with different K_c...
- How does the choice of K_c affect the Bode plot?
- Why should K_c be kept under an upper bound?

Bode stability criterion: Third-order process example

P controller for a FOPTD [cf. Seborg Example 14.4]

Suppose $G_v = G_m = 1$, $G_p = \frac{K_p}{(\tau_p s+1)^3}$, with P controller $G_c = K_c$.
Determine the range of K_c to ensure closed-loop stability.

- Bode criterion: $G_{OL}(i\omega) = \frac{K_c K_p}{(1+\omega^2\tau_p^2)^{3/2}}$, $\angle G_{OL}(i\omega) = -3\arctan\omega\tau_p$.
  1. By definition $\angle G_{OL}(i\omega_c) = -\pi$, we get

$$\omega_c = \tan(\pi/3)/\tau_p = \sqrt{3}/\tau_p$$

  2. To ensure $|G_{OL}(i\omega_c)| < 1$, we need

$$\frac{K_c K_p}{(1+\omega_c^2\tau_p^2)^{3/2}} = \frac{K_c K_p}{(1+3)^{3/2}} = \frac{K_c K_p}{8} < 1$$

$$K_c < 8/K_p$$

- A rule about control in third- or higher-order: K_c can't be too large